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Ammonium Uptake by Rice Roots[1]

III. Electrophysiology

Miao Yuan Wang, Anthony D. M. Glass*, Jon E. Shaff, and Leon V. Kochian

Department of Botany, University of British Columbia, Vancouver, British Columbia, Canada V6T 1Z4 (M.Y.W., A.D.M.G.); and United States Plant, Soil and Nutrition Laboratory, United States Department of Agriculture-Agricultural Research Service, Cornell University, Ithaca, New York 14853 (J.E.S., L.V.K.)

The transmembrane electrical potential differences ($\Delta\Psi$) were measured in epidermal and cortical cells of intact roots of 3-week-old rice (*Oryza sativa* L. cv M202) seedlings grown in 2 or 100 μM NH_4^+ (G2 or G100 plants, respectively). In modified Johnson's nutrient solution containing no nitrogen, $\Delta\Psi$ was in the range of −120 to −140 mV. Introducing NH_4^+ to the bathing medium caused a rapid depolarization. At the steady state, average $\Delta\Psi$ of G2 and G100 plants were −116 and −89 mV, respectively. This depolarization exhibited a biphasic response to external NH_4^+ concentration similar to that reported for $^{13}NH_4^+$ influx isotherms (M.Y. Wang, M.Y. Siddiqi, T.J. Ruth, A.D.M. Glass [1993] Plant Physiol 103: 1259–1267). Plots of membrane depolarization versus $^{13}NH_4^+$ influx were also biphasic, indicating distinct coupling processes for the two transport systems, with a breakpoint between two concentration ranges around 1 mM NH_4^+. The extent of depolarization was also influenced by nitrogen status, which was larger for G2 plants than for G100 plants. Depolarization of $\Delta\Psi$ due to NH_4^+ uptake was eliminated by a protonophore (carboxylcyanide-*m*-chlorophenylhydrazone), inhibitors of ATP synthesis (sodium cyanide plus salicylhydroxamic acid), or an ATPase inhibitor (diethylstilbestrol). The results of these observations are discussed in the context of the mechanisms of NH_4^+ uptake by high- and low-affinity transport systems operating across the plasma membranes of root cells.

Ammonium influx by rice (*Oryza sativa* L. cv M202) roots has been shown to exhibit a biphasic dependence on $[NH_4^+]_o$ (Wang et al., 1991, 1993a, 1993b). At low $[NH_4^+]_o$, influx is mediated by a saturable HATS that exhibits high Q_{10} values between 10 and 30°C and a significant sensitivity to metabolic inhibitors (Wang et al., 1993b). At elevated $[NH_4^+]_o$ (between 1 and 40 mM), NH_4^+ influx increases in a linear fashion with increasing $[NH_4^+]_o$, and, although still exhibiting energy dependence, this LATS was shown to be less responsive to metabolic inhibitors (Wang et al., 1993b). A biphasic pattern of NH_4^+ uptake of this sort, with both saturable and linear phases, was first reported in *Lemna* by Ullrich et al. (1984).

To make a definitive evaluation of the thermodynamics of NH_4^+ influx (passive versus active transport), it is essential to determine the chemical potential difference for NH_4^+ between the cytoplasm and external media and $\Delta\Psi$ across the plasma membrane. In a previous communication (Wang et al., 1993a), we made use of $^{13}NH_4^+$ efflux to estimate cytoplasmic $[NH_4^+]$ by means of compartmental analysis. So far as we are aware, only one report measuring $\Delta\Psi$ in rice roots has appeared in the literature: Usmanov (1979) reported $\Delta\Psi$ to be −160 mV. As early as 1964, Higinbotham et al. noted the marked depolarizing effect of $[NH_4^+]_o$ on coleoptile cell $\Delta\Psi$ in oats. Likewise, Walker et al. (1979a, 1979b) demonstrated the transport of ammonium and methylamine across the plasmalemma of *Chara* and the depolarizing effects of these cations. The most detailed study of the concentration dependence of $\Delta\Psi$ depolarization by NH_4^+ was undertaken by Ullrich et al. (1984) using *Lemna*. Below 0.2 mM $[NH_4^+]_o$ both NH_4^+ uptake and $\Delta\Psi$ depolarization responded in a saturable fashion with half-saturation values of 17 μM for both processes. From 0.2 to 1 mM, net uptake of NH_4^+ responded linearly to $[NH_4^+]_o$, with no further $\Delta\Psi$ depolarization. On the basis of this observation, Ullrich et al. (1984) concluded that the linear system might result from diffusion of NH_4^+ or NH_3 across the plasmalemma.

Therefore, the present study was initiated to estimate $\Delta\Psi$ in intact rice roots under conditions corresponding to those employed to estimate cytoplasmic $[NH_4^+]$ in our previous study and to determine the concentration dependence of the depolarizing effect of $[NH_4^+]_o$. The effects of metabolic inhibitors on $\Delta\Psi$ were also examined.

MATERIALS AND METHODS

Growth of Plants

Rice (*Oryza sativa* L. cv M202) seeds were surface sterilized in 1% NaOCl for 30 min and rinsed with deionized water.

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* Corresponding author; fax 1-604-822-6089.

Abbreviations: CCCP, carboxylcyanide-*m*-chlorophenylhydrazone; CN^-, (sodium) cyanide; DES, diethylstilbestrol; $\Delta\Psi$, transmembrane electrical potential difference; G2 or G100, rice seedlings grown in MJNS containing 2 or 100 μM NH_4Cl, respectively; HATS or LATS, high-affinity or low-affinity NH_4^+ transport systems, respectively; MJNS, modified Johnson's nutrient solution; $[NH_4^+]_o$, external ammonium concentration; *p*CMBS, *p*-chloromercuribenzenesulfonate; SHAM, salicylhydroxamic acid; Q_{10}, ratios of rates at temperatures differing by 10°C.

Seeds were allowed to imbibe overnight in aerated deionized water at 38°C before being planted on plastic mesh mounted on the bottoms of polyethylene cups. Four cups (three to four seeds per cup) were set in the lid of a 1-L black polyethylene vessel with the solution level just above the seeds. Seeds were allowed to germinate in the dark (at 20°C) for 4 d. At d 5, rice seedlings were exposed to light and MJNS containing the designated levels of NH_4Cl. The composition of MJNS, growth conditions, nutrient supply, and pH adjustment were described in a previous report (Wang et al., 1993a). The growth media in the 1-L polyethylene vessels were completely replaced on alternate days and the nutrient levels were topped up with concentrated stock solutions daily. Rice plants used in the experiments were 3-week-old G2 or G100 plants.

Measurements of Cell Membrane Potential

Plasma membrane $\Delta\Psi$ of rice roots were measured as described by Kochian et al. (1989) and Glass et al. (1992). In short, rice plants were secured in the larger part of a flow-through Plexiglas impalement chamber and one intact root was carefully placed over the platinum pins in a narrow section of the chamber. This root was held firmly during the impalement by two short lengths of Tygon tubing, from each of which a small wedge had been cut. The tubing was placed on either side of the impalement zone to clamp the root in place. All impalements were made in a region about 1 to 3 cm behind the root tip, using a hydraulically driven, three-dimensional micromanipulator (model MO-20, Narashige USA, Greenvale, NY). Both the Plexiglas impalement chamber and the micromanipulator were mounted on the microscope stage. Microelectrodes (including impaling, reference, and grounding electrodes) were made from 1.0-mm single-barreled borosilicate glass tubing pulled to a tip diameter of approximately 0.5 μm and filled with 3 M KCl (adjusted to pH 2 to reduce tip potentials). Measured membrane potentials of root cells, which are the voltage differences between the impaling and reference electrodes, were amplified and recorded on a strip chart recorder. During impalement, solutions were continuously delivered from an air-pressured reservoir to the chamber through Tygon tubing at controlled flow rates (approximately 7.5 mL min^{-1}).

Experimental Treatments

At the beginning of each experiment, the impalement was made on G2 or G100 roots bathed in their growth media (MJNS containing 2 or 100 μM NH_4Cl, respectively) and the membrane potential was recorded ($\Delta\Psi_{G2}$ or $\Delta\Psi_{G100}$). MJNS without NH_4^+ is referred to throughout as the −N solution. Before applying each treatment, the −N solution was introduced to obtain a resting membrane potential, $\Delta\Psi_{-N}$, as the point of reference. Roots were allowed to equilibrate for at least 3 to 5 min in this −N solution to reach the resting potential before the subsequent treatment solutions were introduced.

Effect of $[NH_4^+]_o$ on $\Delta\Psi$

Roots were exposed to $[NH_4^+]_o$ of 2, 5, 10, 25, 50, 75, 100, 250, or 500 μM for studying the HATS and 1, 2.5, 5, 10, 20, 30, or 40 mM NH_4Cl for investigating the LATS; all exposures were against a background of MJNS. When roots were exposed to several different $[NH_4^+]_o$ during a single impalement, G2 or G100 medium was flushed through the chamber before each change of NH_4^+ concentration. When $\Delta\Psi$ returned to its original ($\Delta\Psi_{G2}$ or $\Delta\Psi_{G100}$) value, we were satisfied that the physiological status of the root had returned to its original condition.

Effect of Accompanying Anion on $\Delta\Psi$

To evaluate the contribution of the accompanying anion to the observed depolarization of $\Delta\Psi$ by NH_4^+ salts in the low concentration range, $\Delta\Psi$ was measured in the following solutions in sequence: (a) 50 μM $CaCl_2$, (b) 50 μM $CaSO_4$, (c) 100 μM NH_4Cl, (d) 50 μM $(NH_4)_2SO_4$. Likewise, in the high concentration range, $\Delta\Psi$ was measured in (e) 5 mM $CaCl_2$, (f) 5 mM $CaSO_4$, (g) 10 mM NH_4Cl, and then (h) 5 mM $(NH_4)_2SO_4$. These concentrations were chosen to provide equivalent anion charge in all treatments.

Effects of Metabolic Inhibitors on NH_4^+-Induced Depolarization of $\Delta\Psi$

The same metabolic inhibitors used in the $^{13}NH_4^+$ influx study (Wang et al., 1993b) were used to investigate effects on NH_4^+-induced depolarization of $\Delta\Psi$. These included 1 mM NaCN plus 1 mM SHAM, 10 μM CCCP, 50 μM DES, and 1 mM *p*CMBS. This study involved three steps. First, the responses of $\Delta\Psi$ to additions of 0.1 or 10 mM NH_4Cl were determined in sequence. Second, the inhibitor to be evaluated was first introduced in −N solution. When $\Delta\Psi$ had reached a new steady state, this solution was replaced with the inhibitor plus 0.1 or 10 mM NH_4Cl in sequence. Finally, the solution containing inhibitor plus NH_4Cl was replaced by −N solution. When a new steady value of $\Delta\Psi_{-N}$ had been reached, 0.1 and then 10 mM NH_4Cl were added to the −N solution in sequence. The NH_4Cl concentrations, 0.1 or 10 mM in MJNS, were selected as representative levels for the operation of the HATS or the combined HATS and LATS (Wang et al., 1993a).

RESULTS

Transmembrane Electrical Potentials of Rice Roots

Plasma membrane $\Delta\Psi$ for epidermal and cortical cells of 3-week-old rice roots (Table I) were measured in either 0.2 mM $CaSO_4$ alone ($\Delta\Psi_{CaSO4}$), or −N solution, G2, or G100 medium ($\Delta\Psi_{-N}$, $\Delta\Psi_{G2}$, and $\Delta\Psi_{G100}$, respectively). As presented in Table I, $\Delta\Psi_{CaSO4}$ values were consistently more negative than $\Delta\Psi$ measured in other solutions. Likewise, the $\Delta\Psi_{-N}$ values were more negative than the corresponding $\Delta\Psi_{G2}$ or $\Delta\Psi_{G100}$ values. The depolarizing effect of NH_4Cl additions can be directly compared in Table I for a particular root type because −N solution and G2 or G100 medium differed only by the presence of NH_4Cl in MJNS. Therefore,

Table I. *Membrane potentials of 3-week-old rice roots (G2 and G100 plants) measured in different bathing solutions (0.2 mM $CaSO_4$, MJNS −N, MJNS + 2 μM NH_4^+, or MJNS + 100 μM NH_4^+)*

	G2 Plants	G100 Plants
	mV	
$\Delta\Psi_{CaSO_4}$[a]	−140 ± 3.5 (*n* = 5)[d]	−135 ± 1.8 (*n* = 53)
$\Delta\Psi_{-N}$[b]	−129 ± 1.0 (*n* = 184)	−131 ± 0.6 (*n* = 197)
$\Delta\Psi_{G2}$ or $\Delta\Psi_{G100}$[c]	−116 ± 2.1 (*n* = 14)	−89 ± 2.4 (*n* = 28)

[a] G2 or G100 plants were impaled in 0.2 mM $CaSO_4$ solution. [b] G2 or G100 plants were impaled in −N solution. [c] G2 or G100 plants were impaled in MJNS containing either 2 or 100 μM NH_4Cl, respectively. [d] Average value ± SE; *n*, number of observations.

both $\Delta\Psi_{G2}$ and $\Delta\Psi_{G100}$ represented the membrane potentials of root cells adapted to their respective growth conditions.

Contribution of the Accompanying Anions on $\Delta\Psi$

Figure 1 reveals that there was a very small depolarizing effect of Ca^{2+} salts compared with NH_4^+ salts under conditions where the concentration of the accompanying anion was held constant. Also, there was virtually no difference between the depolarizing effects of Cl^- and SO_4^{2-}. This was true as well at the higher concentrations of Ca^{2+} salts and NH_4^+ salts (traces e, f, g, and h in Fig. 1). In the lower concentration range, no repolarization of $\Delta\Psi$ was observed until the Ca^{2+} salts or NH_4^+ salts were withdrawn from the chamber. By contrast, in 5 mM $CaCl_2$ complete repolarization and even hyperpolarization was evident within 10 min of exposure (Fig. 1e). This was not the case for the equivalent sulfate treatment (Fig. 1f). Likewise, at elevated $[NH_4^+]_o$ there was some evidence of repolarization in NH_4Cl (Fig. 1g), but this was only partial. Only after removal of the NH_4^+ salts was complete repolarization observed.

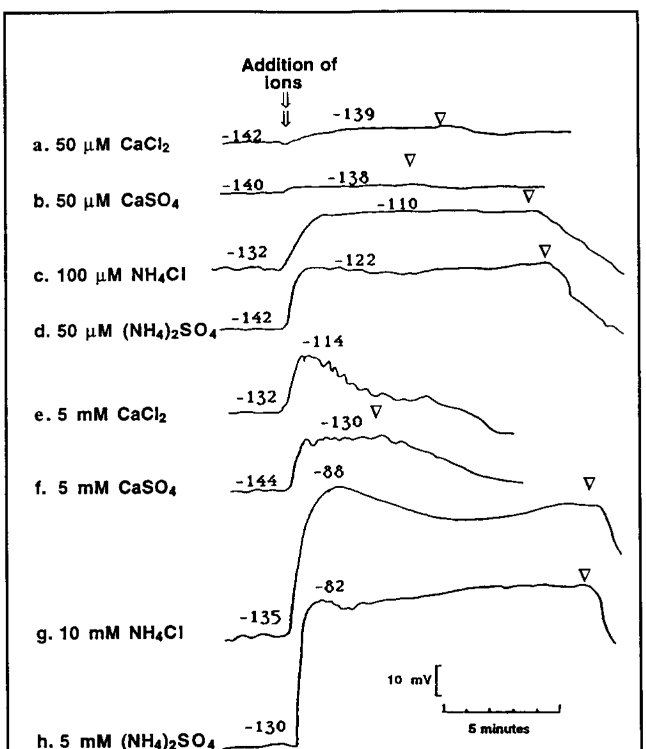

Figure 1. Representative traces to demonstrate the contribution of the accompanying anions to depolarization of $\Delta\Psi$ elicited by exposure of 3-week-old rice roots to different concentrations of various salts. ∇, The salts were withdrawn from MJNS. Each treatment was repeated on three separate plants.

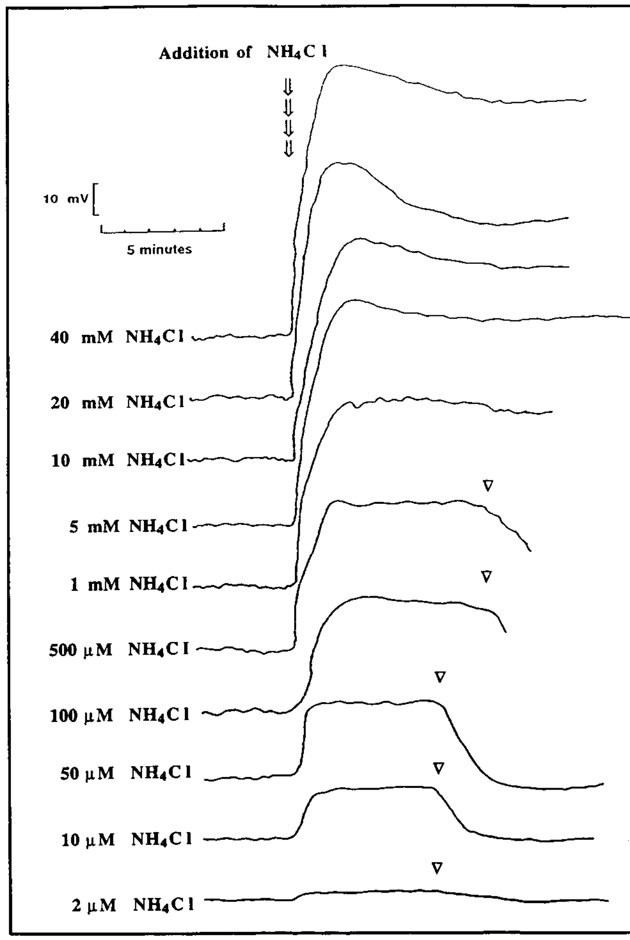

Figure 2. Representative traces from G2 plants showing the depolarization of root cell $\Delta\Psi$ induced by adding various concentrations of NH_4Cl. ∇, NH_4Cl was withdrawn from MJNS.

Effect of $[NH_4Cl]_o$ on $\Delta\Psi$

The addition of NH_4Cl to the −N solution induced a strong depolarization of $\Delta\Psi$ (Fig. 2). This depolarization occurred rapidly after the introduction of NH_4Cl, even at very low concentrations (e.g. 2 μM NH_4Cl). The time required to reach the initial maximum depolarization was from 0.5 to 2 min, increasing with increasing $[NH_4Cl]_o$.

The depolarization of $\Delta\Psi$ was positively correlated with $[NH_4Cl]_o$. A saturable pattern was evident in the range from 2 to 1000 μM NH_4Cl (Fig. 3A) for both G2 and G100 plants.

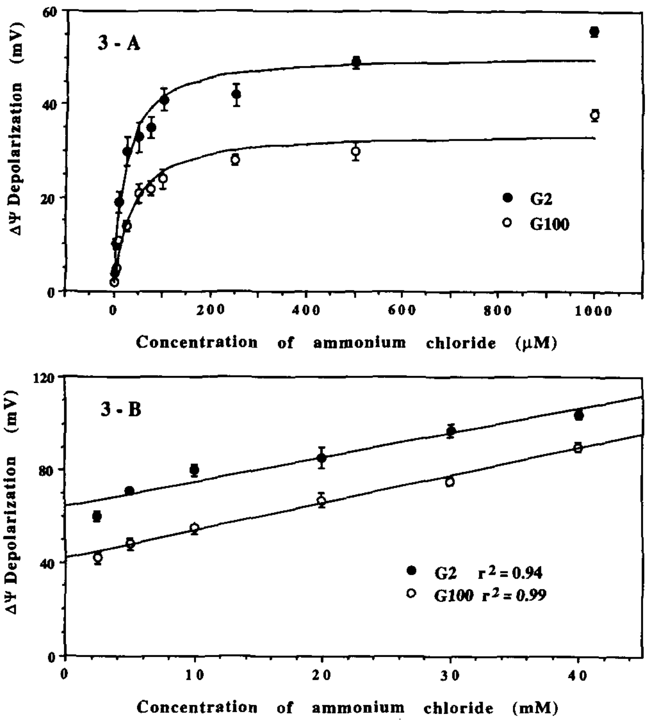

Figure 3. Concentration dependence of net $\Delta\Psi$ depolarization of root cells. Rice seedlings were 3 weeks old and grown in either 100 μM (G100) or 2 μM (G2) NH_4^+. The −N media were used as basal solutions in which to measure the resting $\Delta\Psi$. Each point is the average of three measurements from each of three individual plants. The vertical bar is the SE. A, Low $[NH_4Cl]_o$ range (<1 mM); B, high range (1–40 mM).

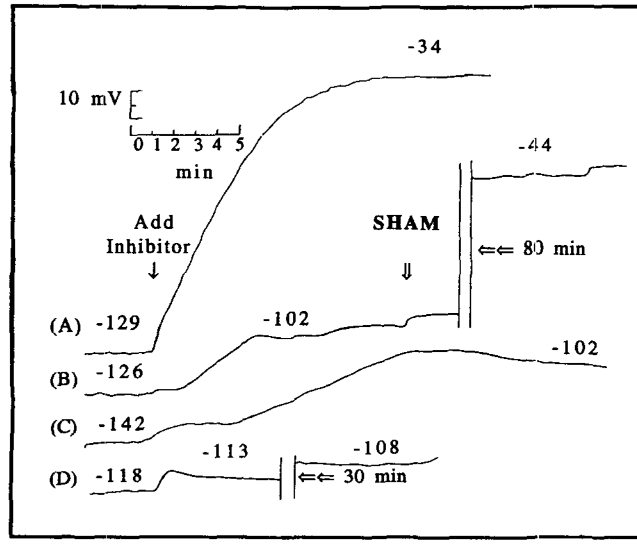

Figure 4. Representative time course traces showing effects of metabolic inhibitors on $\Delta\Psi$. The inhibitors were: A, 10 μM CCCP; B, 1 mM CN^- + SHAM (CN^- was added into −N media alone and then SHAM was added at ⇓); C, 50 μM DES; D, 1 mM *p*CMBS. Each treatment was repeated on at least three individual roots. The space between two bars (||) is the omitted period in minutes.

Estimated half-saturation values for net depolarization (analogous to K_m values) were 21.8 ± 2.7 μM for G2 plants and 35.0 ± 8.0 μM for G100 plants, whereas the maximum depolarization values (analogous to V_{max} values) were 50.6 ± 2.0 mV for G2 plants and 34.3 ± 1.9 mV for G100 plants. Kinetic parameters were obtained by fitting the data to the Michaelis-Menten equation by means of the nonlinear regression computer program Systat (Wilkison, 1987), as used in our earlier kinetic study of $^{13}NH_4^+$ influx (Wang et al., 1993b). Between 1 and 40 mM $[NH_4Cl]_o$ (Fig. 3B), the magnitude of the depolarization increased linearly with increasing concentrations of NH_4Cl. This relationship was observed for both G2 and G100 rice plants, although the extent of depolarization was smaller for the latter.

Effect of Metabolic Inhibitors on $\Delta\Psi$

Figure 4 shows the effects of four metabolic inhibitors on $\Delta\Psi$ recorded in −N solutions. The largest depolarization of $\Delta\Psi$ (95 mV) was induced by the protonophore CCCP, whereas CN^- + SHAM and the ATPase inhibitor DES elicited depolarizations of 82 and 40 mV, respectively. The external protein modifier *p*CMBS caused only a small depolarization (8 mV). Representative traces depicting the effects of each of these inhibitors on NH_4^+-induced depolarization of $\Delta\Psi$ are shown in Figure 5. In Table II, the effects of these inhibitors

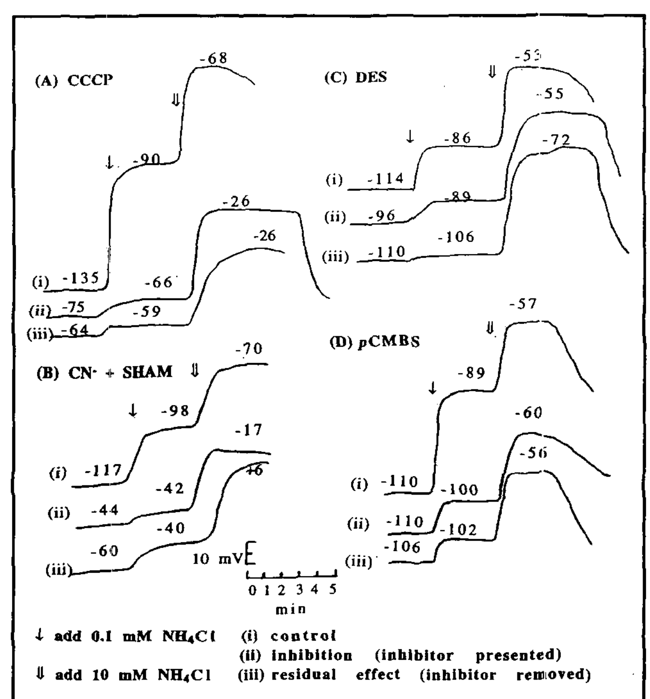

Figure 5. Representative traces for the effects of NH_4Cl on $\Delta\Psi$ depolarization in the presence or absence of metabolic inhibitors in −N media. Metabolic inhibitors were those shown in Figure 4.

Table II. *Effect of metabolic inhibitors on the depolarization of ΔΨ due to NH_4^+ uptake via HATS or LATS in G2 plants*

The inhibitors used were 10 μM CCCP, 1 mM CN^- + 1 mM SHAM, 50 μM DES, and 1 mM *p*CMBS.

	Reduction of ΔΨ Depolarization			
	CCCP	CN^- + SHAM	DES	*p*CMBS
Due to NH_4^+ uptake by HATS[a]				
(i) Control	0	0	0	0
(ii) Plus inhibitor	89	91	72	52
(iii) Residual effect	91	68	81	90
Due to NH_4^+ uptake by LATS[b]				
(i) Control	0	0	0	0
(ii) Plus inhibitor	9	0	14	—[c]
(iii) Residual effect	34	—	3	—

[a] The values of ΔΨ were measured when roots were bathed in MJNS containing 0.1 mM NH_4^+ in the absence (i and iii) and presence (ii) of the inhibitors. The percentage reductions of ΔΨ depolarization were calculated from the differences between control values for ΔΨ induced by NH_4^+ and depolarization values in the presence of the inhibitor (ii) or after removal of the inhibitor (iii). [b] The values of ΔΨ for LATS were the differences between measured ΔΨ at 10 mM (for HATS + LATS) and at 0.1 mM NH_4Cl (for HATS). Then the percentages were calculated as described above (footnote a). [c] The calculated values were negative due to the lower ΔΨ depolarization of the control.

on the NH_4^+-induced depolarization of ΔΨ are expressed as a percentage of the depolarization under the control conditions, i.e. in the absence of the inhibitor. The data are presented as follows: (i) control: in the absence of the inhibitor the reduction of NH_4^+-induced depolarization of ΔΨ was zero; (ii) plus inhibitor: reduction of NH_4^+-induced depolarization of ΔΨ varied from 9 to 91%, depending on the inhibitor used and $[NH_4^+]_o$; and (iii) residual effect: the residual effect after removal of the inhibitor from external solutions on NH_4^+-induced depolarization of ΔΨ. The $[NH_4Cl]_o$ values employed were 0.1 and 10 mM, respectively, chosen to represent the HATS and the combined HATS + LATS. In Table II the depolarizations of ΔΨ caused by 0.1 mM $[NH_4Cl]_o$ were subtracted from those caused by 10 mM $[NH_4Cl]_o$ to represent the effect due to the LATS alone. In the presence of the various inhibitors, the depolarization of ΔΨ induced by the HATS was generally reduced by greater than 50%. By contrast, depolarization of ΔΨ due to NH_4^+ uptake through the LATS was only slightly affected by the presence of inhibitors. Table II also reveals that there was virtually no recovery from the inhibitor treatments following removal of the inhibitors from the external media.

DISCUSSION

Anion Effect

A perennial problem associated with attempts to evaluate the electrical effect of a particular ion is the contribution of the accompanying counterion. This problem has rarely been acknowledged in published studies. However, indirect approaches, such as comparisons of the depolarizing effects of NO_3^- in NO_3^--induced and uninduced plants, have been employed to dissect out the anion effect (Glass et al., 1992). Another approach that has proven effective is to switch from one anion to another [e.g. $CaCl_2$ to $Ca(NO_3)_2$] without changing the accompanying cation or its concentration. As a result, the observed changes of ΔΨ are due solely to the anion effect (McClure et al., 1990; Glass et al., 1992). The results of such studies have demonstrated that NO_3^- can strongly depolarize ΔΨ, and these observations have formed the basis of currently proposed proton/nitrate co-transport mechanisms (Ullrich and Novacky, 1981; McClure et al., 1990; Glass et al., 1992).

In the present study, low concentrations of Cl^- (100 μM) provided in the form of the calcium salt elicited a very small depolarization (Fig. 1a). Replacing this solution with the same concentration of $CaSO_4$ (Fig. 1b) confirmed that Cl^- was responsible for most of this depolarization. Thus, when these calcium salts were replaced by their ammonium equivalents, maintaining the same anion concentration, the significant depolarization of ΔΨ could be attributed largely to NH_4^+. Although the depolarizing effects of the calcium salts present at 5 mM were significantly higher than at 50 μM (Fig. 1, e and f), transfer to the equivalent ammonium salts can be seen to induce a much larger depolarization (53 compared with 18 mV; Fig. 1, g and e). Even though it was not possible to isolate quantitatively the contribution of Cl^- for studies of LATS, we believe that the NH_4^+ effect still predominated, even at high external $[NH_4Cl]_o$. In fact, the difference between traces g and e (Fig. 1) can be attributed to the difference between NH_4^+ and Ca^{2+} effects, since Cl^- was maintained at the same level. Thus, the depolarizations referred to in the remainder of the paper were interpreted as being due predominantly to the transport of NH_4^+.

A feature of these initial studies was the apparent repolarization of ΔΨ following depolarization in the chloride solutions (Fig. 1, e and g) at high $[Cl^-]_o$. Although repolarization to the resting potential was not complete in 10 mM NH_4Cl, the extent of the initial repolarization was comparable to that in $CaCl_2$, where repolarization was completed. A similar spontaneous repolarization of ΔΨ was noted in *Lemna* and in barley roots following depolarization of ΔΨ by NO_3^- (Ullrich and Novacky, 1981; Glass et al., 1992).

Depolarization of ΔΨ by HATS and LATS

Addition of ammonium chloride into −N solutions induced a rapid depolarization of membrane potential of rice epidermal and cortical cells (Figs. 1 and 2). This was evident even at very low concentration (2 μM NH_4Cl) (Fig. 2). Ullrich et al. (1984) reported that addition of NH_4^+ immediately decreased the membrane potentials of *Lemna gibba*. Likewise, the ΔΨ of green thallus cells of *Riccia fluitans* were rapidly depolarized by $[NH_4Cl]$ levels as low as 1 μM (Felle, 1980). As can be seen from Figure 2, the time to reach initial maximum depolarization increased from approximately 0.5 to 3 min with increasing concentrations of NH_4Cl.

The depolarization of ΔΨ by NH_4^+ exhibited a biphasic concentration dependence (Fig. 3, A and B), similar to the effect on NH_4^+ influx into roots of rice (Wang et al., 1993b).

In the low concentration range (<1 mM), depolarization of the membrane potential saturated in response to $[NH_4^+]_o$ (Fig. 3A). Both net flux and unidirectional influx of NH_4^+ in rice roots have been shown to respond to $[NH_4^+]_o$ in a similar fashion (Youngdahl et al., 1982; Wang et al., 1991, 1993b). Estimated half-saturation values for NH_4^+-induced depolarization (analogous to K_m values) were 21.8 ± 2.7 μM for G2 plants and 35.0 ± 8.0 μM for G100 plants. These values were somewhat lower than the K_m values for $^{13}NH_4^+$ influx, 32 μM and 90 μM, respectively (Wang et al., 1993b). Since our studies were undertaken with the same rice variety as employed for the $^{13}NH_4^+$ influx experiments, these differences may arise from the fact that there were differences in growth conditions for plants used for the two studies or that membrane depolarization reflects the net, rather than the unidirectional, effect of ion fluxes. Another factor, already addressed above, is the possible effect of the accompanying anions. The maximum depolarizations (analogous to V_{max} values) were 50.6 ± 2.0 mV and 34.3 ± 1.9 mV for G2 and G100 plants, respectively. The larger depolarizing effects of $[NH_4^+]_o$ in G2 compared with G100 plants (Fig. 3, A and B) correspond to the higher values of $^{13}NH_4^+$ influx observed in G2 compared with G100 plants (Wang et al., 1993b). Clearly the depolarization of $\Delta\Psi$ in response to $[NH_4^+]_o$ (<1 mM) was due to the carrier-mediated NH_4^+ uptake that exhibited Michaelis-Menten kinetics (Wang et al., 1993b). Similar saturable patterns of $\Delta\Psi$ depolarization were associated with the uptake of either NH_4^+ or NO_3^- in *Lemna* (Ullrich and Novacky, 1981; Ullrich et al., 1984) and the uptake of both NH_4^+ and $CH_3NH_3^+$ in cells of *R. fluitans* (Felle, 1980).

Between 1 and 40 mM, the depolarization of $\Delta\Psi$ increased linearly with increasing $[NH_4Cl]_o$ (Fig. 3B) in a manner similar to that observed for $^{13}NH_4^+$ influx (Wang et al., 1993b). Both G2 and G100 rice plants exhibited this linear response, but the extent of depolarization was smaller in G100 plants, where $^{13}NH_4^+$ influx was also smaller. The concentration-dependent data for depolarization of $\Delta\Psi$ by LATS was fitted by linear regression with r^2 values of 0.94 and 0.99 for G2 and G100 rice plants, respectively. A similar linear response to $[NH_4^+]_o$ was reported for net NH_4^+ uptake by *Lemna* at $[NH_4^+]_o$ values between 0.1 and 1 mM (Ullrich et al., 1984). However, in this concentration range, NH_4^+ uptake by *Lemna* was not associated with further depolarization of $\Delta\Psi$. Ullrich et al. (1984) interpreted this pattern as due to a diffusive uptake of NH_4^+ or NH_3. It is clear that NH_3 influx would not depolarize $\Delta\Psi$. However, it is not clear how NH_4^+ uptake could occur without further $\Delta\Psi$ depolarization, unless NH_4^+ influx was associated with a stoichiometric anion influx or cation efflux resulting in an electroneutral transport.

To better understand the relationship between NH_4^+ uptake and changes in $\Delta\Psi$, the observed values of $\Delta\Psi$ depolarization were paired with the data for $^{13}NH_4^+$ influx from Wang et al. (1993b) at each $[NH_4^+]_o$ (Fig. 6). It is evident that the depolarization of $\Delta\Psi$ was strongly correlated with $^{13}NH_4^+$ influx and that the relationship was biphasic. By use of a computer-based procedure to determine objectively the "breakpoints" for the biphasic pattern (Rygiewicz et al., 1984), the correlation coefficient established a breakpoint at 1 mM $[NH_4^+]_o$. The biphasic pattern (Fig. 6) indicates that NH_4^+ influx and the depolarization of $\Delta\Psi$ are due to two distinct systems for NH_4^+ uptake by rice roots, i.e. a HATS and a LATS. The larger slope of the lines for the low concentration range for G2 and G100 plants suggests that the HATS is more electrogenic than the LATS. This may be due to the increasingly electroneutral NH_4^+ transport at high $[NH_4Cl]_o$.

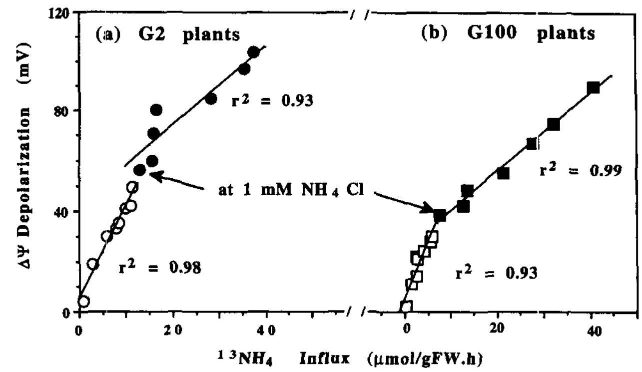

Figure 6. The relationship between $^{13}NH_4^+$ influx (data from Wang et al., 1993b) and net depolarization of $\Delta\Psi$ at the same $[NH_4^+]_o$ for G2 and G100 plants.

In the present study, the electrophysiological evidence suggested that at high $[NH_4Cl]_o$ ammonium is taken up by rice roots in the cation form (NH_4^+) despite the presence of a relatively high concentration of NH_3 in solution. Alternatively, it might be argued that depolarization of $\Delta\Psi$ may be due to the inhibition of the H^+-ATPase by NH_3 at high $[NH_4^+]_o$. However, the lack of a pronounced increase of ^{13}N uptake at pH values approaching the pK_a for NH_4^+ does not support this interpretation (Wang et al., 1993b). In addition, the rapid repolarization of $\Delta\Psi$ following removal of external NH_4^+ (Fig. 1, traces g and h) is unexpected considering that the half-time for cytoplasmic ^{13}N exchange is approximately 7 min (Wang et al., 1993a).

Calculation of the Free Energy for NH_4^+ Transport

The average $\Delta\Psi$ values were substantially more negative in G2 plants impaled in 2 μM NH_4^+ than in G100 plants impaled in 100 μM NH_4^+ (Table I). Furthermore, the extent of the depolarization of $\Delta\Psi$ by NH_4^+ was consistently greater for G2 plants than for G100 plants at a particular $[NH_4^+]_o$. The average $\Delta\Psi$ value was −116 mV for G2 plants and −89 mV for G100 plants (Table I). For both G2 and G100 plants, the resting potentials in the absence of NH_4^+ ($\Delta\Psi_{-N}$) were in the range of −120 to −140 mV. In low-salt bathing medium (0.2 mM $CaSO_4$), the transmembrane electrical potentials ($\Delta\Psi_{0.2\,mM}$ CaSO4) were 25 mV more negative than $\Delta\Psi_{-N}$ and 45 mV more negative than $\Delta\Psi_{G2}$ and $\Delta\Psi_{G100}$. These differences reflect the contributions to the membrane depolarization from the various ions present in MJNS. Since the values of $\Delta\Psi_{-N}$ and of $\Delta\Psi_{G2}$ and $\Delta\Psi_{G100}$ were measured in the same basal medium (MJNS), the observed differences must be due largely to the $[NH_4^+]_o$ in the bathing medium.

The measured $\Delta\Psi$, together with values for cytoplasmic $[NH_4^+]$, are needed to estimate the electrochemical potential difference for NH_4^+ across the plasmalemma, which in turn allows us to determine the energy requirement for transport

(Findlay and Hope, 1976). Taking 3.72 mM and 20.55 mM as cytoplasmic [NH_4^+] and −116 mV and −89 mV as steady-state $\Delta\Psi$ for G2 and G100 roots, respectively (Wang et al., 1993a), at a series of given [NH_4^+]$_o$, the Nernst potentials were estimated for G2 and G100 roots. From these values, the free energy required to transport NH_4^+ across the plasmalemma can be computed from the differences between measured membrane potentials ($\Delta\Psi_{G2}$ or $\Delta\Psi_{G100}$) and estimated Nernst potentials at specific [NH_4^+]$_o$ (Fig. 7). The estimated free energy differences for NH_4^+ distribution were positive at or below 42 μM for G2 and 655 μM for G100 plants (Fig. 7). This means that below these concentrations, NH_4^+ uptake by G2 and G100 roots, respectively, must be active (Fig. 7). These concentrations represent the lower limits for active transport under steady-state conditions. However, displacing [NH_4^+]$_o$ to values greater than 2 or 100 μM, respectively, will elevate the limit for active transport because of further $\Delta\Psi$ depolarization and increased cytoplasmic [NH_4^+]. Above these minimum levels, the uptake of NH_4^+ may occur via passive transport systems, down the electrochemical potential gradients for NH_4^+.

As pointed out previously (Wang et al., 1993b), these free energy estimations provide only a prediction of the feasibility of the uptake process occurring under the prescribed conditions. For both G2 and G100 plants, the predicted [NH_4^+]$_o$ for the shift from active to passive uptake was considerably lower than the breakpoint determined by the kinetics analyses (42 and 655 μM versus 1 mM). Thus, one must be cautious in identifying a specific transport system based purely on thermodynamic or kinetic considerations.

Mechanisms of NH_4^+ Uptake by HATS and LATS

The preceding section has demonstrated that at low [NH_4^+]$_o$ (<42 μM for G2 plants and <655 μM for G100 plants), NH_4^+ influx appears to be an active process in roots of rice plants. However, the details of this mechanism are unknown for rice and for any higher plants. Possible mechanisms for this active uptake via HATS include: (a) a proton:NH_4^+ symport; and (b) a specific NH_4^+ ATPase. The results of the inhibitor studies, both for the electrical potentials in the

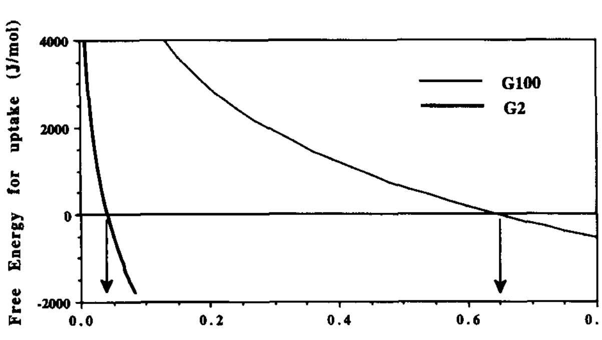

Figure 7. Free energy requirement for NH_4^+ uptake as a function of external [NH_4^+]. Values of cytoplasmic [NH_4^+] were taken from our previous study (Wang et al., 1993a). Arrows indicate the [NH_4^+]$_o$ below which NH_4^+ uptake is against the electrochemical potential gradient for G2 and G100 plants, respectively.

present study and for $^{13}NH_4^+$ influx (Wang et al., 1993b), provide evidence for a dependence (either direct or indirect) on the proton motive force. Application of CCCP caused 89 and 85% inhibition, respectively, of membrane depolarization by NH_4^+ and $^{13}NH_4^+$ influx in solutions containing 100 μM NH_4^+. The strong inhibitory effects of CN^- + SHAM on depolarization of $\Delta\Psi$ (91%) and on $^{13}NH_4^+$ influx (81%) confirm the dependence of these processes on a source of metabolic energy without distinguishing the nature of the mechanisms. The effects of DES, an inhibitor of the H^+-ATPase, indicated the involvement of the proton pump, suggesting that H^+ transport might be involved.

The results of the present and earlier studies (Wang et al., 1993b) strongly suggest that the two systems, HATS and LATS, have different mechanisms of energy coupling. Above 42 μM for G2 and 655 μM for G100 plants, NH_4^+ transport was predicted to be a passive process. This prediction is borne out by the generally smaller effects of metabolic inhibitors at high external [NH_4^+] than at low [NH_4^+]$_o$ (Wang et al., 1993b; present study), although $^{13}NH_4^+$ influx showed greater sensitivity to inhibitors than the $\Delta\Psi$ depolarization. There is virtually no information available regarding the energy coupling for the LATS. Passive entry of NH_4^+ might occur via an electrogenic uniport (Kleiner, 1981; Ullrich et al., 1984). This may be a specific channel for NH_4^+ or a shared cation channel. For example, the recently described K^+ channel in *Arabidopsis* has been shown to have an NH_4^+ conductance that is approximately 30% of the K^+ conductance (Schachtman et al., 1992). Also, in the cyanobacterium *Anabaena variabilis* (Avery et al., 1992), the uptake of Cs^+ (a K^+ analog at the uptake step) and of NH_4^+ were closely related. Thus, low-affinity NH_4^+ transport might occur via the K^+ channel.

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